

Q: Is there a positive relationship between birth rates and restrictions on civil liberties across countries?

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$$\text{Birth Rate} = 16.9423 + 0.2294 \times (\text{SIGI Score}) + -0.0002 \times (\text{GDP per capita})$$

Ho: There is not a significant relationship between birth rates and restrictions on civil liberties

Ha: There is a positive relationship between birth rate and restrictions on civil liberties



Every one point increase in SIGI score (restrictions on civil liberties) is estimated to increase birth rate by 0.2294 P value =0.000

Before Control Variable

	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.9640	1.535	5.189	0.000	4.929	10.999
Q("OBS_VALUE")	0.3633	0.046	7.881	0.000	0.272	0.454

After Controlling (GDP)

	coef	std err	t	P> t	[0.025	0.975]
Intercept	16.9423	1.749	9.688	0.000	13.484	20.401
Q("OBS_VALUE")	0.2294	0.043	5.386	0.000	0.145	0.314
Q("GDP per capita")	-0.0002	2.73e-05	-7.663	0.000	-0.000	-0.000

0.3633 - 0.2294 =
(0.1339) : represents the
portion of error made, and
explained by adding a
control variable. Adding the
control variable isolates the
relationship more clearly.

R-squared: 0.314
Adj. R-squared: 0.308
F-statistic: 62.11
Prob (F-statistic): 9.36e-13
Log-Likelihood: -492.84
AIC: 989.7
BIC: 995.5

R-squared: 0.522
Adj. R-squared: 0.515
F-statistic: 73.27
Prob (F-statistic): 3.17e-22
Log-Likelihood: -464.86
AIC: 935.7
BIC: 944.5

The change in the
Adjusted R-Squared tells
us that GDP can explain a
lot of the variation in our
dependent variable.

The P-Value of the
F-statistic got bigger
after adding control, yet
it is still extremely small,
showing overwhelming
evidence against our null
hypothesis

Regression Results